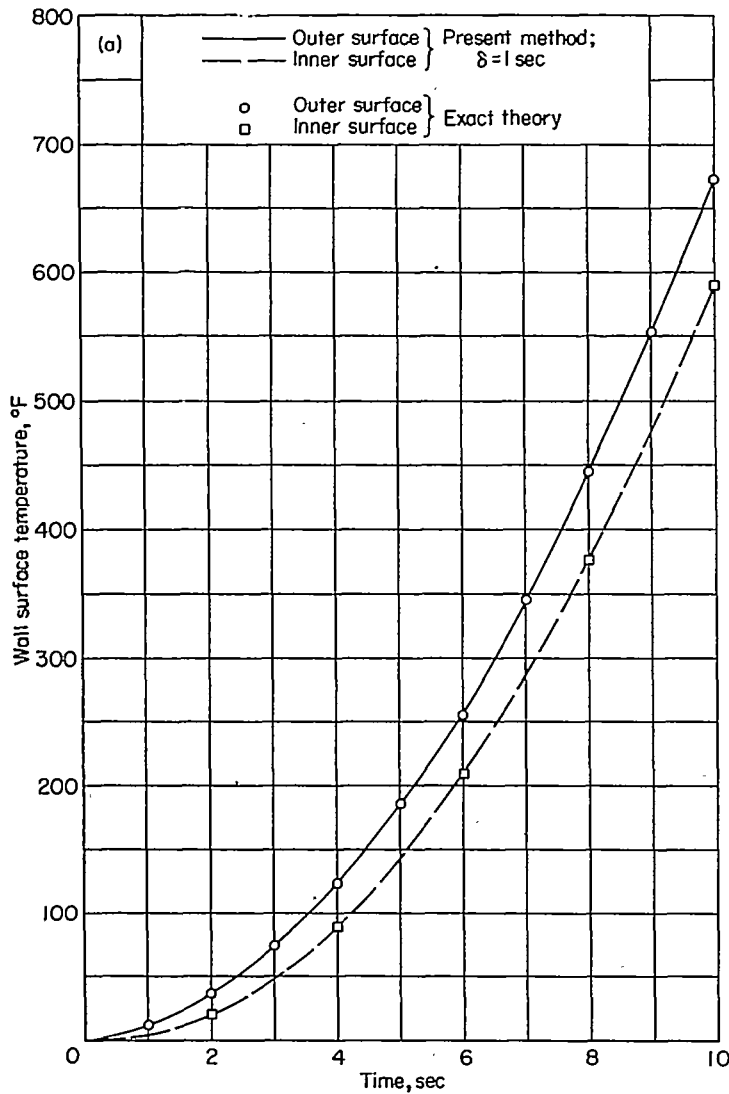
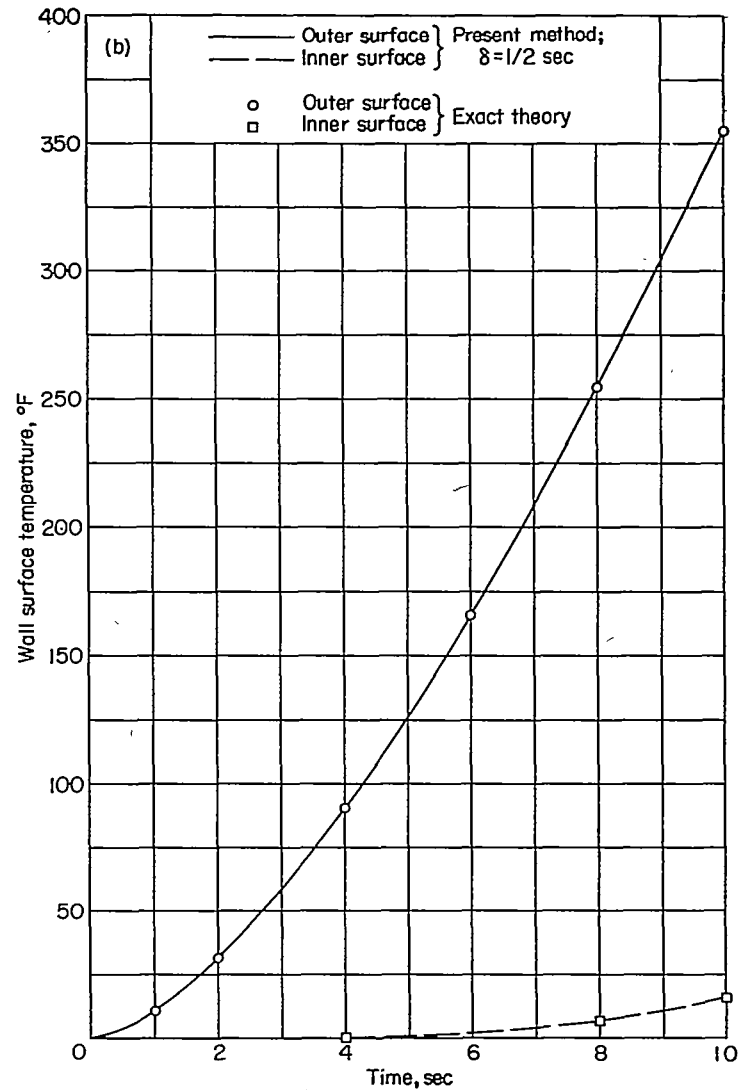




(a) 1/2-inch-thick wall.



(b) 3-inch-thick wall.

FIGURE 1.—Example 1. Temperatures of copper-wall surfaces. Adiabatic-wall temperature varies linearly from 0° to 10,000° in 10 seconds; $h=100$ Btu/(hr) (sq ft) (°F).

FIGURE 1.—Concluded.

by the present method, the maximum difference between methods being about 1°.

Problem for example 1 (b).—The conditions for example 1 (b) are the same as those for example 1 (a), except that the copper wall is 3 inches thick, or $l=3/4$ foot.

Solution for example 1 (b).—Since this wall is so thick, more highly transient conditions prevail throughout the heating period. A value of δ of $1/2$ second or $1/7200$ hour was therefore used. With the same procedure as used before, the results of the present method are given as continuous lines in figure 1 (b), while the results from equation (44) are given by the symbols. The agreement is evident.

EXAMPLE 2

Problem.—Example 2 illustrates the principal advantage of the present method; that is, its capability of handling arbitrary variations of T_{aw} and h . Neither is it necessary to know a mathematical formulation for these variations.

A 1/2-inch copper wall which is initially at a temperature

of zero is aerodynamically heated on one side and insulated on the other. The time histories of T_{aw} and h are given at $1/2$ -second intervals in the following table:

Time, sec	T_{aw} , °F	h , Btu/(hr) (sq ft) (°F)	Time, sec	T_{aw} , °F	h , Btu/(hr) (sq ft) (°F)
0	0	36	5.5	5,391	68.4
.5	1,365	41.4	6.0	5,356	66.6
1.0	2,485	45.0	6.5	5,255	63.9
1.5	3,388	48.6	7.0	5,107	60.0
2.0	4,094	52.2	7.5	4,831	55.8
2.5	4,620	55.8	8.0	4,335	52.2
3.0	4,932	60.0	8.5	3,654	48.6
3.5	5,119	63.9	9.0	2,769	45.0
4.0	5,263	66.6	9.5	1,658	41.4
4.5	5,345	68.4	10.0	297.5	36.0
5.0	5,387	69.0			

The temperature history of both wall surfaces is to be found.

Solution.—The heating conditions are severe and continuously transient, with the boundary-layer temperature

rising and falling over $5,000^\circ$ in 10 seconds. A computing interval δ smaller than that used in the first problem is therefore used. Let $\delta = \frac{1}{2}$ second $= 1/7200$ hour. The resulting wall-surface-temperature curves are drawn in figure 2. The circles and squares give the surface temperatures computed by a theoretically exact procedure. Comparison shows the present method to be accurate.

EXAMPLE 3

Problem.—Example 3 is the same as example 2 except that the wall is 3 inches thick and the effective boundary-layer temperatures are slightly different. The initial value of T_{aw} is 0. The subsequent values of T_{aw} are given at 0.5-second intervals by the following time series: $T_{aw} = 1,365, 2,484, 3,386, 4,088, 4,609, 4,915, 5,094, 5,227, 5,296, 5,325, 5,315, 5,265, 5,149, 4,986, 4,694, 4,184, 3,489, 2,591, 1,471, 100$. Find the temperature history of both wall surfaces.

Solution (a) (thick-wall solution).—The value of δ was taken as $\frac{1}{2}$ second. By following the procedure previously

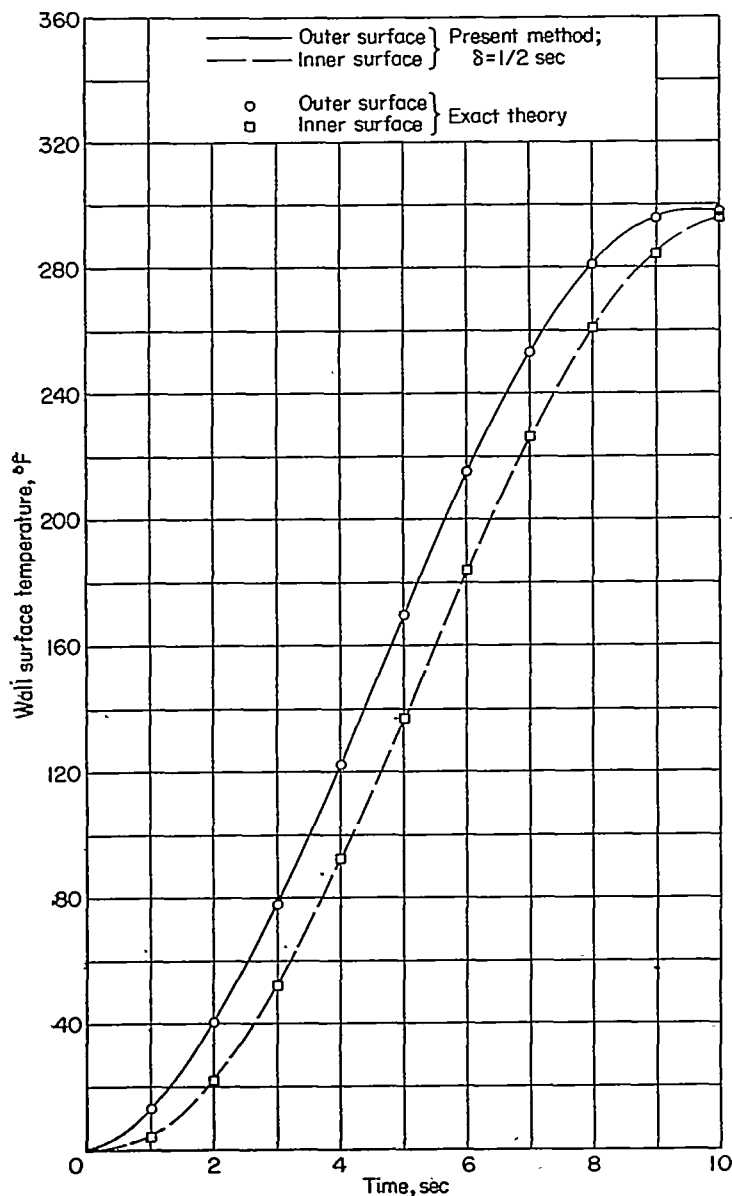


FIGURE 2.—Example 2. Temperatures of $\frac{1}{2}$ -inch copper wall heated according to assigned history of h and T_{aw} .

presented, the wall-surface-temperature curves shown in figure 3 were obtained. Again the symbols give the results of exact theory.

It should be noted that the "exact solution" for examples 2 and 3 is not actually an alternate method of solution for any practical problem but gives a solution to the particular problems only. The "solution" was obtained by working in reverse; that is, a heat flow was assumed and the corresponding boundary-layer characteristics were computed. A truly alternate method of solution is now considered, however.

Solution (b) (infinitely thick-wall solution).—Since the thermal lag of a 3-inch copper wall is so great when subjected to the rapid heating specified by this problem, it appears reasonable to obtain the heated-surface temperature by assuming that the wall is infinitely thick. The memory coefficients are the same for all infinitely thick walls and are given by equations (31). The same temperature formulas, the same values of T_{aw} , and the same values of h are used as before, but the values of H_∞ are given by equation (29). The results calculated by this method are listed along with

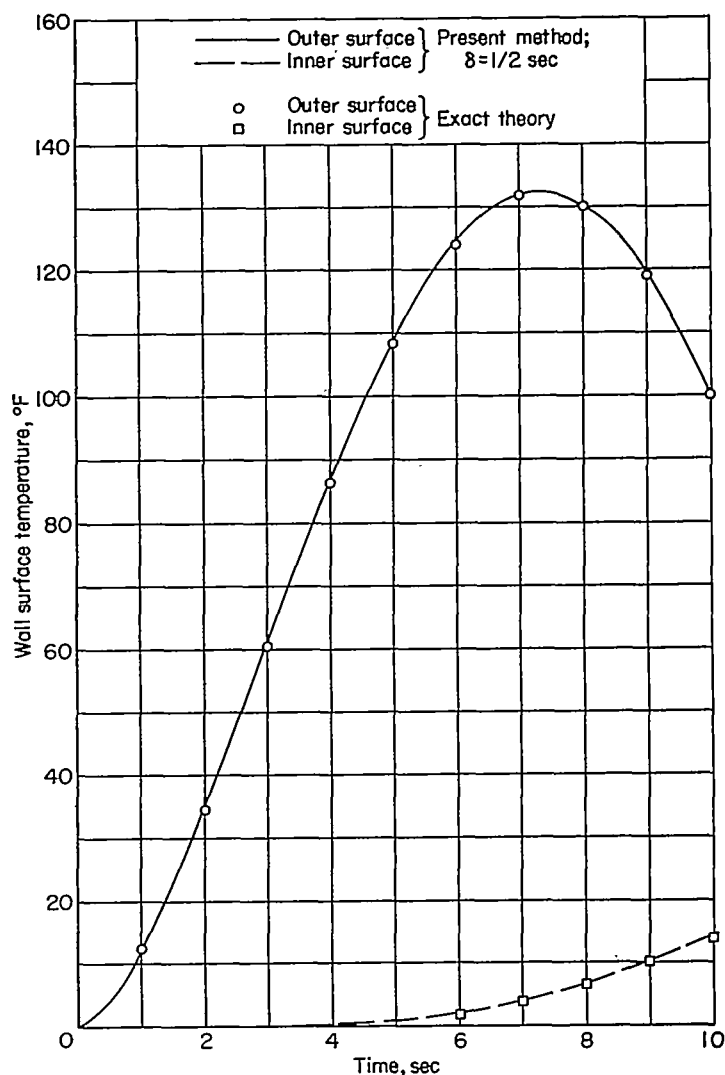


FIGURE 3.—Example 3. Temperatures of 3-inch copper wall heated according to assigned history of h and T_{aw} .

those from the thick-wall solution (a).

Time, sec	T, °F, calculated by—	
	Thick-wall solution	Infinitely thick-wall solution
1	12.8	12.8
2	35.3	35.3
3	61.3	61.3
4	87.0	87.0
5	109.1	109.1
6	124.8	124.8
7	132.3	132.3
8	130.6	130.6
9	119.5	119.4
10	100.0	99.8

The consistency of the alternate methods for a thermally thick wall is evident. The reason for the close agreement may be found in figure 3, which shows that the unheated surface of the 3-inch wall rose to only 14° F.

EXAMPLE 4

Problem.—The most severe test of the present method would occur if there were a large instantaneous increase of T_{aw} . While this condition could hardly happen in flight, it might happen if a research model were suddenly immersed in a high-stagnation-temperature jet. Let a $\frac{1}{2}$ -inch copper wall, initially at a temperature of zero, be instantly subjected to an effective boundary-layer temperature of 5,000° F on one surface while no heat transfer occurs on the other surface. The heat-transfer coefficient is 100 Btu/(hr)(sq ft) (°F). Solve for the temperature history of both wall surfaces for 10 seconds.

Solution.—In this case not only is there a very high transient-temperature condition initially but the instantaneous increase in T_{aw} does not lend itself to approximation by the unit triangle. The simplest procedure is to take small steps for the first few seconds to minimize the errors introduced. In order to help circumvent the difficulty of calculation, an excellent method of approximating the wall surface temperature for the first or first few small steps is to use the following formula from page 109 of reference 5, which gives the temperature on the surface of an infinitely thick wall for a constant flow of heat at the surface:

$$T = \frac{2hT_{aw}}{K} \sqrt{\frac{kt}{\pi}} \quad (45)$$

The values of δ used were $\delta = \frac{1}{8}$ second for 2 seconds, then $\delta = \frac{1}{2}$ second for 8 seconds. Since the use of an equation based on an infinite wall is permissible for a $\frac{1}{2}$ -inch copper wall for at least $\frac{1}{4}$ second, the values of T were computed by equation (45) for the first two $\frac{1}{8}$ -second steps, then by the usual equations. The results are presented in figure 4. Since the inside temperature T_i depends on the outside temperature T and not directly on T_{aw} , there is no particular difficulty of approximation in obtaining T_i . Accordingly, in obtaining T_i , $\frac{1}{8}$ -second steps were taken for 1 second to

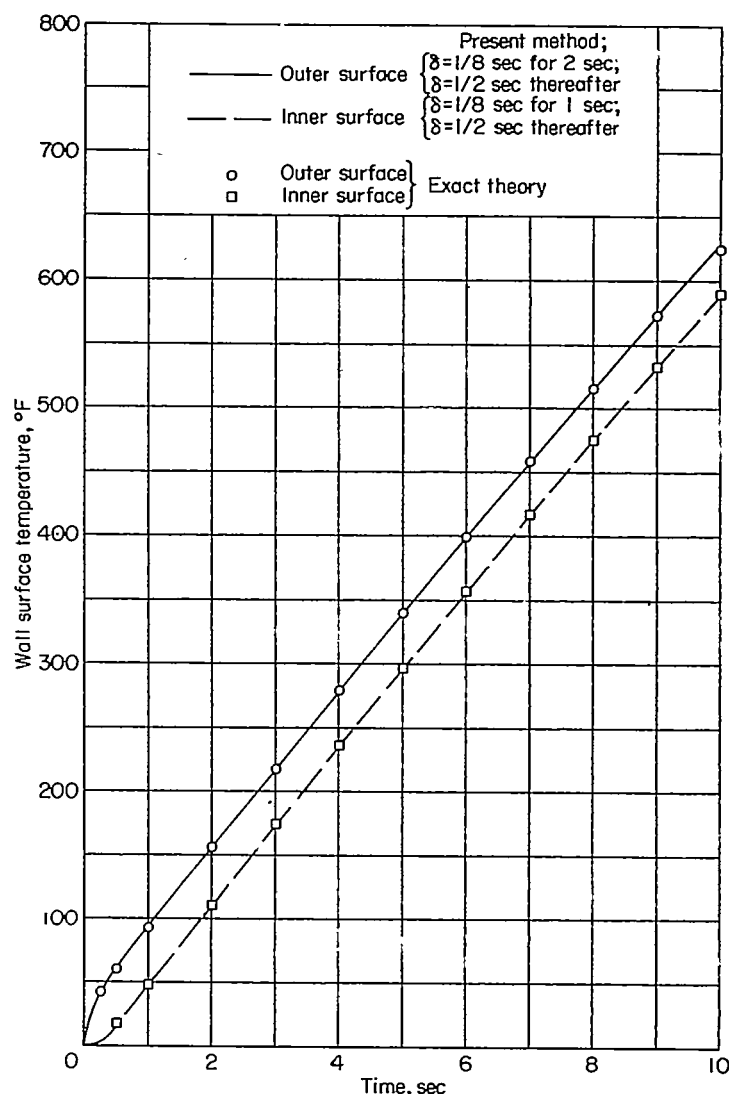


FIGURE 4.—Example 4. Temperatures of $\frac{1}{2}$ -inch copper wall after application of 5,000° F jump in gas temperature. $h=100$ Btu/(hr)(sq ft) (°F).

define the highly transient part of the curve, then $\frac{1}{2}$ -second steps for the remaining 9 seconds.

A theoretically exact solution to this problem was obtained by equation (B7) of appendix B. The results of applying this equation are given by the symbols in figure 4, which shows that agreement was obtained.

EXAMPLE 5

Problem.—A $\frac{1}{8}$ -inch Inconel wall is heated by a high-temperature jet. If $T_{aw}=5,000$ ° F and $h=50$ Btu/(hr)(sq ft) (°F), determine the skin-temperature history for 15 seconds. Neglect radiation.

Solution.—The heat capacity of a $\frac{1}{8}$ -inch Inconel wall was assumed to be $G=0.3229$ Btu/(sq ft) (°F). The example was worked three times with values of δ of 1, 2, and 5 seconds to show how sensitive the "thin wall" formula is to the time interval. Substituting the given constants into equation (34) yielded the results given by the symbols in figure 5. For this example the exact theory is shown by the solid line. Evidently, large time intervals are permissible with this formula.